gpeyt1-03 · USCS from gradation

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A sieve analysis on a split-spoon sample gives: 96% passing No. 4, 3% passing No. 200, D10 = 0.12 mm, D30 = 0.3 mm, D60 = 0.95 mm. The sample is non-plastic, brown, moist, with a natural water content of 9%. The boring log also notes trace mica.

Most nearly, Cu, Cc and the USCS group symbol are:

A. Cu = 7.9, Cc = 1.27, SW
B. Cu = 7.9, Cc = 0.79, SP-SM
C. Cu = 7.9, Cc = 0.79, SP
D. Cu = 7.9, Cc = 0.79, SW

Given equations / data

> Cu = D60/D10; Cc = D30²/(D10·D60); SW requires Cu ≥ 6 and 1 ≤ Cc ≤ 3; fines < 5%

Worked solution

1. Cu = 0.95/0.12 = 7.92 (≥ 6 OK)
2. Cc = 0.3²/(0.12×0.95) = 0.789 (< 1, fails)
3. 96% passes No. 4 → sand; 3% fines < 5% → clean sand
4. Fails Cc → poorly graded sand, SP

Final answer

Cu = 7.9, Cc = 0.79, SP

Why the other values are wrong

- **Cu = 7.9, Cc = 1.27, SW** — inverted Cc
- **Cu = 7.9, Cc = 0.79, SP-SM** — treated 3% fines as a dual-symbol (5–12%) soil
- **Cu = 7.9, Cc = 0.79, SW** — checked Cu only

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.7.2 USCS, p. 110–117; §3.8.4 Gradation Tests, p. 136

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=DJaHlgyaOAg (no creator values copied).

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